Making Peace Between Mortal Enemies: Running a Database Management System in the Linux Kernel

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Montagues vs. Capulets. Bloods vs. Crips. Ginger vs. Mary Ann. For generations, the factions of these famous conflicts could never resolve. We aim to solve one of these: the database management system (DBMS) vs. operating system (OS) feud has gone on for long enough.

1 BACKGROUND

eBPF entered the Linux kernel a decade ago. This modern Linux subsystem enables developers to write safe, event-driven programs that run in kernel-space. We have already demonstrated eBPF's benefits for ancillary DBMS features like training data collection [3] and proxies [4]. However, no one has enabled eBPF programs to store data in an embedded DBMS.

Frankly, OS folks would not take kindly to the idea of a DBMS in the kernel. Both sides of the feud have been entrenched for decades. Mike Stonebraker stated "that [OS] services in many existing systems are either too slow or inappropriate" [6] for DBMSs. PostgreSQL developers leave comments in their code describing Linux's memory accounting as "unbelievably stupid, but the kernel hackers seem uninterested in improving it" [1]. Linus Torvalds is on the record describing "database people" as "seldom hav[ing] any shred of taste" [2]. Despite these scathes, there is hope. With the world so divided, we see an opportunity to bring these two communities together by embedding a DBMS into the Linux kernel.

2 THE CHALLENGE

Sure, I know what you're thinking: people have tried to put a DBMS in the kernel before, and it didn't work out. They failed because their approaches required modifying the OS through modules or compiling a custom kernel, potentially sacrificing stability and security [5]. Instead, we wrote this embedded DBMS *in eBPF*. But eBPF's constraints challenge traditional DBMS design:

- How to store your database contents without heap allocations?
- How to coordinate mutual exclusion with limited atomic primitives?
- How to create locks without acquiring multiple latches?
- How to implement a concurrency control protocol like 2PL, OCC, or MVTO without spinning indefinitely or `yield()`ing?
- How to you persist database contents that are stored in kernel memory?

3 THE RESULT

We had to balance the concerns of both parties: (1) keep the OS people happy by not trashing the kernel or sacrificing OS responsiveness, and (2) keeping the DBMS people happy by offering scalable performance and correctness.

In this talk, we will present BPF-DB, an ACID-compliant, multi-versioned DBMS capable of millions of multi-statement, serializable transactions per second—running entirely in kernel-space. BPF-DB runs without loading a kernel module or compiling a custom kernel, and offers 10x the throughput of Redis.

Let us tell the whole story, share the horrors of eBPF development encountered along the way, and start the healing between DBMS and OS folks by inviting us to HPTS 2024.

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AUTHOR BIO

Matt Butrovich is a final-year Ph.D. candidate in the Computer Science Department, School of Computer Science, at Carnegie Mellon University with a research interest in DBMSs. More specifically, he is exploring operating system and DBMS codesign opportunities via safe kernel extension mechanisms like eBPF. He is extremely fortunate to be advised by Andy Pavlo, and he is a member of the Database Group (CMU-DB), Parallel Data Lab (PDL), and Systems, Networking, and Performance (SNAP) Lab.

Matt previously completed a M.Sc. in Computer Science at CMU, graduating in December 2018. Prior to graduate school, he worked at Western Digital for 8 years performing firmware validation on storage devices while earning a B.Sc. in Computer Science & Engineering at UC Irvine and Irvine Valley College.

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